

DEBATECRAFT CURRICULUM

Bioethics: Science and the Future of Medicine

Authors

Oliver Choi

Mentors

Adrian Chan

Dowan Kim

Summer 2026

Life and Cells

A cell stepped on her sister's toe. The sister said, "Ouch, mitosis."

— Anonymous

1 What is life?

Biology is the scientific study of life — but what exactly is life? It sounds like a strange question, yet scientists have failed to universally conclude on the definition of living. Around 2,400 years ago in Ancient Greece, Aristotle proposed that life was defined by the capacity for growth, nutrition, and reproduction. Later, in the 19th century, Charles Darwin's theory of evolution suggested a new criterion for life: that all living things must reproduce with variations, and that those who adapt best survive and pass on their traits. In the modern era, the development of molecular biology has complicated the question. If we artificially create a cell, should that cell be considered alive?

Rather than settle on one perfect definition, in elementary biology, we will identify life by a set of eight shared characteristics: 1) reproduction, 2) metabolism, 3) homeostasis, 4) heredity, 5) response to stimuli, 6) growth and development, 7) adaptation through evolution, and 8) cellular organization. Living things are called organisms (the adjective is biotic; non-living things are abiotic). Most organisms show all eight characteristics, although there are puzzling cases such as viruses.

The question "what is life?" also extends beyond biology, not because everyone defines biological life differently, but because people often confuse this question with a different one: "Is this living thing a person with moral rights?" The first question is biological, while the second is ethical, legal, and political. This confusion matters because political debates about abortion, end-of-life care, and organ donation can shape laws that affect daily life. In this textbook, we will acknowledge that these questions are interconnected in society, but separate in nature.

In this chapter, we will first learn the eight characteristics of life. Then, we will use those characteristics to examine clear examples of living and non-living things, as well as difficult borderline cases.

1.1 Reproduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si.

1.2 Another placeholder part

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore.

2 Inside the Cell

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

2.1 Placeholder part

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio.

DNA and Genes

1 Molecule of Heredity

2 From DNA to Expressed Traits

3 Mutations

Inheritance

1 Mitosis and Meiosis

2 Mendel's Rules

3 Natural Selection

4 Genetic Disorders

Designing Experiments

1 Examples

1.1 Hypothesis

1.2 Independent and Dependent Variables

1.3 Control Variables

1.4 Sample Size

2 Correlation and Causation

3 Bias and Fairness

4 Reading: Critical Thinking

Introduction to Bioethics

1 idk yet

History of Bioethics

1 Nuremberg

2 Tuskegee

3 Henrietta Lacks

Modern Bioethics

1 CRISPR

2 Gene Therapy

3 AI in Healthcare

Project Design